

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)**ACADEX PRACTICE EXAMINATION****O-LEVEL · BIOLOGY****BIOLOGY 5008/2****PAPER 2 November 2024****2 hours****Candidate Name:** _____**Centre Number:** _____ **Candidate Number:** _____**TIME ALLOCATION & INSTRUCTIONS TO CANDIDATES**

Electronic calculators may be used. Answer all questions in Section A.

Section A: Compulsory structured questions. Section B: Structured essay questions. Write in clear, technical scientific English.

SECTION A

1. A plant cell was placed in a concentrated sucrose solution for 30 minutes. (a) State the process by which water left the cell. (b) Describe the appearance of the cell after 30 minutes. (c) Explain why an animal cell in the same solution behaves differently. *[Cell Biology & Osmosis]* [6]
- (a) State the process by which water moves out of the cell. [1]
- (b) Describe the condition of the plant cell after 30 minutes. [2]
- (c) Explain the difference in response between a plant cell and an animal red blood cell in this solution. [3]
2. (a) Write the balanced word equation for photosynthesis. (b) State two environmental factors that increase the rate of transpiration in maize plants. *[Photosynthesis & Transpiration]* [6]
- (a) State the word equation for photosynthesis. [2]
- (b) Name two external factors that accelerate transpiration. [2]
- (c) Explain why wilting occurs on hot, dry afternoons. [2]